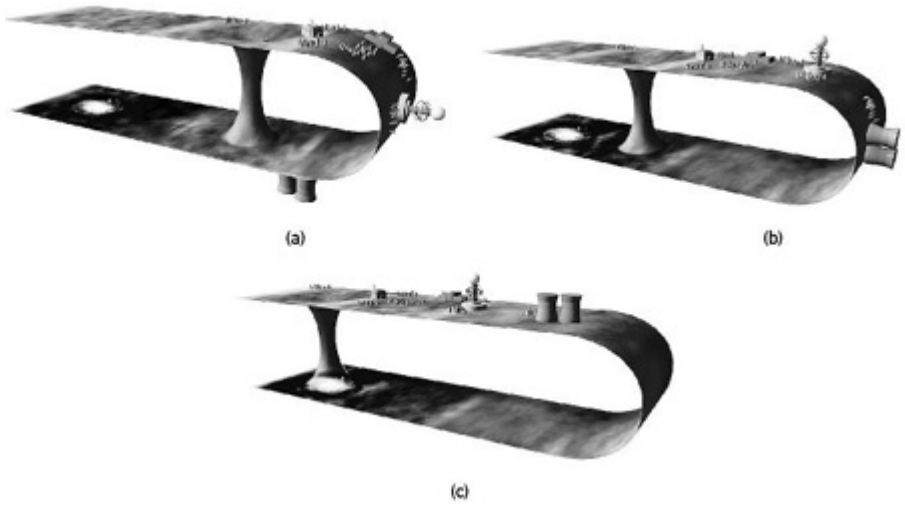


Figure 15.4 (a) A wormhole connecting the Kwik-E-Mart and the nuclear power plant. (b) The lower wormhole opening transported (from the nuclear power plant) to outer space (on spaceship, not shown). The wormhole length remains fixed. (c) The wormhole opening arrives at the Andromeda galaxy; the other opening is still at the Kwik-E-Mart. The length of the wormhole is unchanged throughout the entire voyage.

the ship reaches Andromeda, Lisa tells Bart to pipe down so she can take in the view without disturbance. She's exasperated by his insistence on quickly grabbing the takeout at the Fish Finger Flythrough and heading back to Springfield, but agrees to keep on chatting until he returns. Four hours and a few dozen rounds of tic-tac-toe later, Bart safely sets his ship down on the lawn of Springfield High.

When he looks out the ship window, though, Bart gets a bit of a shock. The buildings look completely different, and the scoreboard floating high above the rollerball stadium gives a date some 6 million years after his departure. "Dude!?!" he says to himself, but a moment later it all becomes clear. Special relativity, he remembers from a heart-to-heart he'd recently had with Sideshow Bob, ensures that the faster you travel the slower your clock ticks. If you travel out into space at high speed and then return, only a few hours might have elapsed aboard your ship while thousands or millions of years, if not more, will have elapsed according to someone stationary. With a quick calculation, Bart confirms that at the speed he was traveling, eight hours elapsed on the ship would mean 6 million years elapsed on earth. The date on the scoreboard is right; Bart realizes he has traveled far into earth's future.